MIL-R-46617A(MU)
5 February 1965
SUPERSEDES
MIL-R-46617(Ord)
3 January 1962

MILITARY SPECIFICATION

REMOVER, AIRCRAFT CANOPY, M3A1, ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M3A1, Assembly, loaded with Cartridge, Aircraft Canopy Remover, M31A1 for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
- MS-28741 - Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command
- F8593701 - Remover, Aircraft Canopy, M3A1, Assembly
- F8797385 - Ammunition Container, Fiber M321 for Remover, Aircraft Canopy, M3A1
- F8797386 - Box, Packing, Ammunition for Remover, Aircraft Canopy, M3A1
- IEL-8593701 - Index to Inspection Equipment List for Remover, Aircraft Canopy, M3A1, Assembly

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:28Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Material, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Torque.- Torque requirements shall be as specified in Table I. No loosening or deformation shall occur upon application of the breakaway torque.

TABLE I

Component	B Seating or Bottoming (Inch pounds)	C Breakaway (Inch pounds)
Block and inside tube after insertion of cartridge	400 + 50	300 min.
Ring, stop and outside tube	500 + 50	400 min.

3.3 Assembly

3.3.1 The body face surface of the plug shall be in complete contact with the cartridge.

3.3.2 The block shall be free to rotate 360 degrees.

3.4 Residual magnetism.- The assembled remover shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.5 Functional.

3.5.1 Ballistics.- Removers loaded with accepted M31A1 cartridges shall propel a 300 pound mass vertically and shall comply with the requirements listed in Table II.

TABLE II

Temperature	Velocity (See 6.3)(min.)	Thrust (min.)	Ignition Delay (See 6.2) (max.)
-65° to -70°F	20 fps	2000 lbs.	40 milliseconds
65° to 75°F	20.5 fps	2600 lbs.	40 milliseconds
160° to 165°F	22.5 fps	3200 lbs.	40 milliseconds

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

3.5.2 The remover shall not fail to fire.

3.5.3 The remover shall function without mechanical failure (See 6.1).

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, in referenced specifications and the following:

3.6.1 Metal defects.- The component shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 Burr.- The component shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The component or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-414, as applicable.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

performed on a defect basis in accordance with classification of defects contained herein, using the sampling plan and associated criteria in Table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Aircraft Canopy Remover, M31A1, Dwg. D-8593735 covering a detail of Dwg. F-8593701.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" Ring missing	Visual
4. Sealing disc missing, damaged or improperly secured	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. "O" Ring split, cut or cracked	Visual
102. Foreign matter	Visual
MINOR	
201. Case scratched, wrinkled, bulged or dented	Visual
202. Varnish missing, or inadequate	Visual

4.2.2.2 Seal, Dwg. C-8593665 covering a detail of Drawing F8593701.

CRITICAL

1. None defined

MAJOR

101. None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MINOR	
201. Inside diameter under flange	Gage
202. Inside diameter adjacent to flange	Gage
203. Radius missing or incorrect	Visual
204. Metal defective	Visual
205. Protective finish inadequate or defective	Visual
206. Burr	Visual
207. Foreign matter	Visual

4.2.2.3 Assembly, Dwg. F-8593701, less all drawings except C-8593644, B-8593654, C-8593657, A-8593663, C-8593709, D-8593714, C-8593715 and AN6227-11.

CRITICAL	
1. Gas port hole obstructed	Visual
2. Any part missing	Visual
3. Improper assembly	Visual
MAJOR	
101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Dwg. F-8593701, less Dwgs. B-8593660, C-8593665, D-8593703, C-8593707 and C-8593711.

Defect	Method of Inspection
CRITICAL	
1. Body face surface of plug not in complete contact with cartridge	Gage
2. Cartridge missing	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 202640-22T06:28Z
Check the source to verify that this is the current version before use.

4.2.2.5 Assembly, Dwg. F-8593701, less all drawings except B-8593660, D-8593703, C-8593707 and C-8593711.

	Defect	Method of Inspection
CRITICAL		
1.	Seal on bearing tube missing	Visual
MAJOR		
101.	Burr	Visual
102.	Foreign matter	Visual
MINOR		
	None defined	

4.2.2.6 Assembly, Dwg. F-8593701 (with stop ring not engaged with outside tube).

	Defect	Method of Inspection
CRITICAL		
1.	Inside Tube not free to slide	Manual
MAJOR		
	None defined	
MINOR		
	None defined	

4.2.2.7 Assembly, Dwg. F-8593701

	Defect	Method of Inspection
CRITICAL		
1.	Mixed removers	Visual
2.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Diameter of block mounting hole	Gage
102.	Diameter of cap mounting hole	Gage
103.	Block not free to rotate counter-clockwise	Manual
104.	Burr	Visual
105.	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MINOR	
201. Length between mounting holes	Gage
202. Total length, max.	Gage
203. Location of labels	Visual
204. Shipping plug missing	Visual
205. Seal missing or insecure	Visual
206. Protective finish inadequate or defective	Visual

4.2.2.8 Packing in Fiber Container, Dwg. F-8797385

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.9 Packing in Wooden Box, Dwg. F-8797386

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping, missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Breakaway torque.- The breakaway torque shall be applied to each threaded joint listed in 3.2. Failure of any remover to comply with the requirements of 3.2, Table I shall be cause for rejection of the remover.

4.2.3.2 Residual magnetism.- Each remover shall be tested for excessive residual magnetism. Failure of any remover to comply with the requirements of 3.4 shall be cause for rejection of the remover.

4.2.3.3 Functional.- The following tests shall be performed by the Government.

4.2.3.3.1 Ballistics.- Fourteen units shall be tested to determine compliance with the requirements of 3.5.1, 3.5.2 and 3.5.3. The units shall be temperature conditioned as provided in Table IV and 4.3.3.1.1. The variables sampling plan of MIL-STD-414, example C-2 or C-3, as applicable for the AQL listed in Table IV shall be used for each sample of 10 removers. Failure of any remover to comply with the requirements of 3.5.1, 3.5.2 and 3.5.3 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

TABLE IV

Temperature	Sample Size	A.Q.L.
-65° to -70°F	2	0.65%
65° to 75°F	10	0.65%
160° to 165°F	2	0.65%

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:28Z
Check the source to verify that this is the current version before use.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593701.

4.3 Test methods and procedures

4.3.1 Breakaway torque.- A torque wrench, set at the torque requirement specified in 3.2, shall be used to determine the adequacy of the joint. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 Residual magnetism.- The remover shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the remover 6 inches from and at the same height as the compass, in a north-south horizontal position (block end at south) with the compass set in an east-west heading as shown in figure 1. The remover shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

4.3.3 Functional

4.3.3.1 Ballistics.- The remover shall be rigidly supported in the fixture by the mounting points and fired with an M3A1 Initiator connected to the remover with 10 feet of aircraft hose, size 4, as specified in MS-28741. The remover shall comply with the requirements of 3.5. The remover shall be tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the remover shall be reconditioned at the original temperature.

4.3.3.1.1 Temperature conditioning.- A standard M3A1 Remover, loaded with a dummy M31A1 Cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A. - Packing and marking shall be in accordance with Drawings F-8797385 and F-8797386.

5.2 Data cards.- Data card information shall be as specified in MIL-A-2550.

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:28Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 "Ignition delay" is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise on the thrust time record.

6.3 Velocity applies at tube separation.

Custodian:

Army - MU

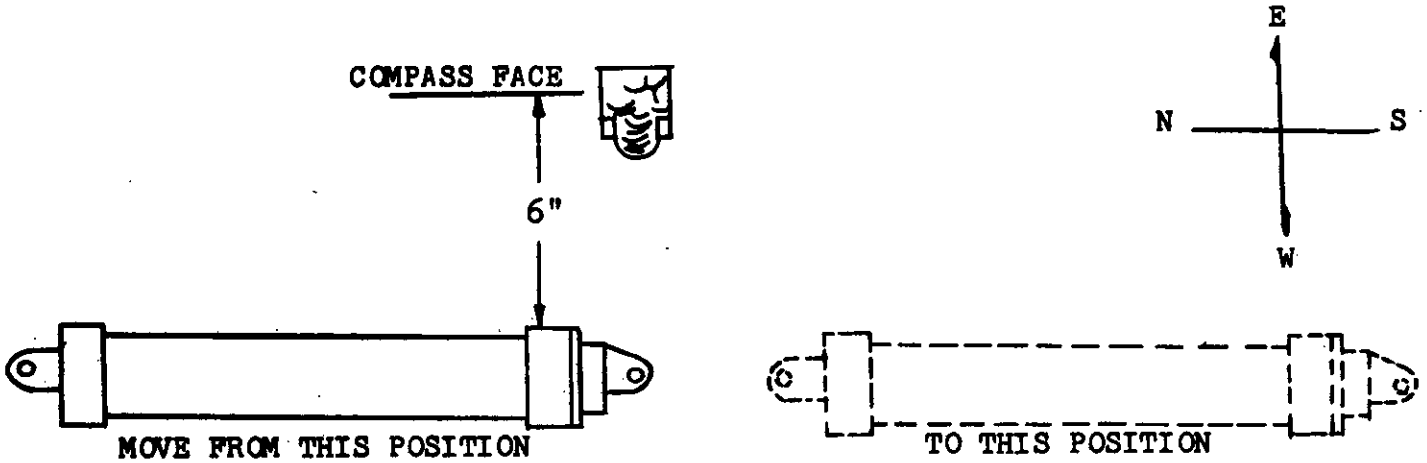
Preparing activity:

Army - MU

Project No. 1375-A853

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

PILOT'S STANDBY COMPASS, TYPE AN-5766



PLAN VIEW

NOTE:

Remover and compass shall be in a horizontal plane

Figure 1

Layout for determination of residual magnetism in Remover, Aircraft Canopy, M3A1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 44 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:28Z
Check the source to verify that this is the current version before use.